

Answer the following questions

Question 1: Choose the correct answer

(25 Marks)

- 1) Why do we need biological neural networks?
 - a) To solve tasks like machine vision & natural language processing.
 - b) To apply heuristic search methods to find solutions of problem.
 - c) To make smart human interactive & user-friendly system.
 - d) All of the mentioned.
- 2) Which of the following is TRUE about Perceptrons?
 - a) It is an artificial neuron and is the fundamental unit of a neural network.
 - b) It is also called single layer neural network or single layer binary linear classifier.
 - c) A neural network is a composition of perceptrons connected in different ways.
 - d) All of the above.
- 3) Which of the following is FALSE about activation functions?
 - a) Activation functions help in achieving non-linearity in deep neural networks.
 - b) These are also called squashing functions as these squash the output under a certain range.
 - c) Commonly used activation functions are step, sigmoid, tanh, ReLU and softmax.
 - d) None of the above.
- 4) Output of step (threshold) activation function ranges from:
 - a) Either 0 or 1
 - b) 0 to 1
 - c) -1 to 1
 - d) Either -1 or 1
- 5) Which of the following is TRUE about Softmax and Sigmoid function?
 - a) Softmax is usually used for hidden layers and sigmoid for outer layers.
 - b) Sigmoid is usually used for hidden layers and softmax for outer layers.
 - c) Softmax function is usually used for binary classification problem.
 - d) All of the above.
- 6) In a neural network, synaptic weights represent:
 - a) Dendrite branching
 - b) Connection strengths between neurons
 - c) Activation functions
 - d) Bias values
- 7) What is supervised learning?
 - a) weight adjustment based on deviation of desired output from actual output
 - b) weight adjustment based on desired output only
 - c) weight adjustment based on actual output only
 - d) None of these
- 8) a multilayer feed forward neural network has 5 input neurons, a first hidden layer with 4 neurons, a second hidden layer with 3 neurons, and 2 output neurons. How many parameters does this hidden layer have (including the bias parameters)?
 - a) 42
 - b) 38
 - c) 47
 - d) 32
- 9) In a three layers network, number of classes is determined by?

- a) number of neurons in second layer
b) number of neurons in third layer

- c) number of neurons in second and third
d) None of these

10) If I am using all features of my dataset and I achieve 99 % accuracy (error = 0) on my training set, but ~96 % accuracy on testing set, what should I look out for?

- a) Underfitting (b) Nothing, the model is perfect c) Overfitting d) None of these

A three-layer neural network is being trained through a backpropagation learning algorithm with a learning rate of 0.8. The input layer has two neurons N_1 and N_2 with inputs 2 and -1.5, respectively. The hidden layer has two neurons N_3 and N_4 and the output layer has a single neuron N_5 . Neurons N_3 , N_4 and N_5 employ a hyperbolic tangent activation function each. The desired response of the network (d) is 0.5. At a particular iteration of the algorithm, the weights are set as :

$w_{13} = w_{14} = 0.7$, $w_{23} = w_{24} = 0.6$, $w_{35} = w_{45} = 1$, $w_{03} = 0.4$, $w_{04} = -1$, $w_{05} = 1.2$

Solve the questions from (Questions 11 to 25)

11) the neuron N_3 has an activation value $y_3 = \dots\dots\dots$

- a) 0.3 (b) 0.9 c) 0.6 d) None of these

12) the neuron N_4 has an activation value $y_4 = \dots\dots\dots$

- a) -0.3 b) -0.2 (c) -0.5 d) None of these

13) the neuron N_5 has an activation value $y_5 = \dots\dots\dots$

- a) 1.854 (b) 1.454 c) 2.462 d) None of these

14) the neuron N_5 has an output value $f(y_5) = \dots\dots\dots$

- a) 0.586 b) 0.455 (c) 0.897 d) None of these

15) Calculate the gradient of w_{05} when the error value ($e = d - f(y_5)$) of the neuron N_5 is

- (-0.396) (a) 0.078 b) 0.019 c) 0.003 d) 0.316 e) None of these

16) Calculate the gradient of w_{35} when the neuron N_3 has an output value $f(y_3) = 0.716$

- a) 0.228 b) 0.347 (c) 0.056 d) 0.01 e) None of these

17) Calculate the gradient of w_{45} when the neuron N_4 has an output value $f(y_4) = -0.462$

- a) -0.084 b) -0.011 c) -0.059 (d) -0.036 e) None of these

18) Calculate the gradient of w_{13} when the gradient of $w_{03} = 0.038$

- a) 0.024 (b) 0.076 c) 0.050 d) 0.046 e) None of these

19) Calculate the gradient of w_{23} when the gradient of $w_{03} = 0.038$

- a) -0.015 b) -0.036 (c) -0.057 d) -0.077 e) None of these

20) Calculate the gradient of w_{14} when the gradient of $w_{04} = 0.061$

- (a) 0.122 b) 0.045 c) 0.312 d) 0.376 e) None of these

21) Calculate the gradient of w_{24} when the gradient of $w_{04} = 0.061$

a) -0.044 b) -0.024 c) -0.067 ~~d) -0.092~~ e) None of these

22) Calculate the new weight value of w_{03} for the next iteration.

a) 0.583 ~~b) 0.370~~ c) 0.876 d) 0.501 e) None of these

23) Calculate the new weight value of w_{04} for the next iteration.

a) -2.103 b) -0.470 c) -2.076 ~~d) -1.049~~ e) None of these

24) Calculate the new weight value of w_{05} for the next iteration.

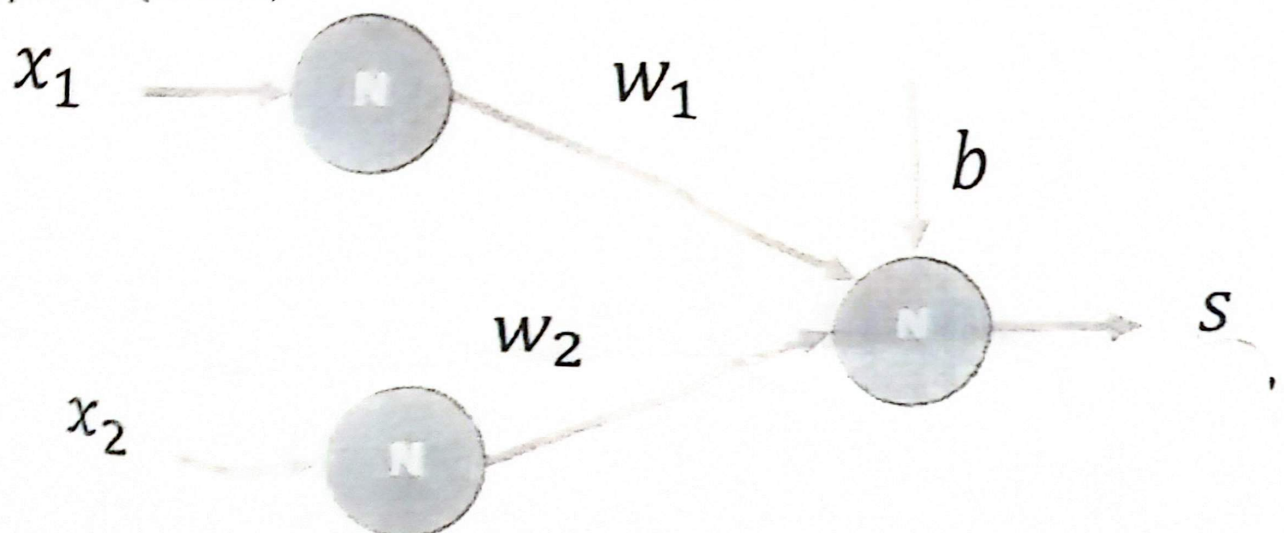
a) 2.546 b) 2.143 ~~c) 1.138~~ d) 1.954 e) None of these

25) Calculate the new weight value of w_{23} for the next iteration.

~~a) 0.646~~ b) 0.345 c) 1.521 d) 0.831 e) None of these

Question 2:

1) The following perceptron receives two inputs (x_1, x_2) and produces a single binary output s . (6 Marks)



a. Find the appropriate values of the weights (w_1, w_2, b) to make the perceptron ...